James Clements III

Portland, Oregon - james.l.clements.iii@gmail.com - (503) 555-0142
github.com / clementsj - clementsj.dev

Summary

Senior audio DSP engineer with seven years of shipped real-time C++ on embedded ARM and desktop platforms. Particular focus on partitioned convolution reverberation, room correction, and low-latency scheduling. Comfortable across the full audio stack from kernel-level driver glue to plug-in UI in JUCE.

Experience

Cantus Audio, Portland OR — *Lead audio engineer*, 2022 – present.

- Own the real-time engine in *Cantus Verb Pro*: a 96 kHz partitioned convolution reverb shipping on macOS, Windows, and an in-house ARM hardware unit (Cortex-A78).
- Brought worst-case 64-sample block latency from 11 ms to 2.8 ms by redesigning the FFT block scheduler around lock-free SPSC queues.
- Mentor three junior engineers and run the team's weekly DSP reading group.

CCRMA, Stanford CA — *Research engineer*, 2019 – 2022.

- Core contributor to the *faust2* compiler; landed the SSA-based delay-line optimiser used in versions 2.50 onward.
- Co-authored two ICASSP papers on physically modelled string synthesis.
- Built and maintained the lab's nightly regression suite (1300 tests).

Smule, San Francisco CA — *Audio engineer intern*, summer 2018.

- Implemented the pitch-correction path for *Smule Sing!* on iOS, using a phase-vocoder front end and an autocorrelation-based pitch tracker.

Education

Stanford University — M.S., Music, Science & Technology (CCRMA), 2019. Thesis: *Sparse Time-Varying Filters for Physically-Modelled Strings.*
Oberlin Conservatory + College — B.A. Computer Science, B.Mus. Harp Performance (dual-degree program), 2016. *Magna cum laude.*

Technical skills

Languages — C, C++17 (daily), Python 3 (daily), TypeScript, Rust, Faust.
DSP — Partitioned convolution, FFT-based filtering, phase vocoders, FDN reverb, physical modelling, room correction, psychoacoustic masking.
Audio frameworks — JUCE, CoreAudio, ASIO, ALSA, VST3, AU, AAX, WebAudio.
Tooling — CMake, Bazel, perf, Tracy, ASan, TSan, UBSan, Ableton Live, Pro Tools.
Other — Concert-level pedal harp; conversational French; amateur radio (KC7QHK).

Selected publications & projects

- Clements, J. *"Sparse, Time-Varying All-Pole Models for Plucked Strings,"* ICASSP 2022.
- Clements, J. and Smith III, J. O. *"A Lazy Scheduler for Faust Block Diagrams,"* ICASSP 2021.
- **abcjsharp** — fork of `abcjs` with proper harp grand-staff rendering. github.com/clementsj/abcjsharp.
- **mdlout** — the Markdown-to-PDF pipeline used to build this CV. *References available on request.*